CRYPTOGRAPHIC ENGINEERING MANUSCRIPT (RELEASE V2.0) · ARCHIVAL RECORD: CERN ZENODO DOI 10.5281/zenodo.23510005

Marma-Crypto (Marma-256): Bijective Vector Commitments and Constant-Rate Traitor Tracing for PAdES-B-LTA Signatures via Overdetermined Planar Incidence Lattices over $\mathrm{GF}(2^{257})$ (DSTU 4145-2002)

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Abstract

Long-term archival electronic signatures under ETSI EN 319 142-1 (PAdES-B-LTA) and ETSI EN 319 122-1 (CAdES-X-Long) suffer from a structural *container-stream decoupling vulnerability*: the cryptographic signature value resides in an external or detachable ASN.1 DER `/Contents` dictionary or `.p7s` container, whereas the human-inspected visual appearance stream (`/AP`) embedded in the PDF/A or SVG page description is an unauthenticated Euclidean graphic carrying zero bits of cryptographic entropy. When enterprise Document Management Systems (DMS) flatten PDF files, strip detached `.p7s` bundles, or experience incremental shadow updates, the visual trust badge survives intact while cryptographic verifiability is either lost or silently decoupled from the rendered glyph coordinates.

We introduce **Marma-Crypto (`Marma-256`)**, a deterministic, bijective geometric vector commitment and traitor-tracing protocol that embeds a full 256-bit post-combiner elliptic curve signature digest plus an orthogonal 10-bit leak-attribution canary directly into the IEEE-754 double-precision (`binary64`) floating-point vertex coordinates of an overdetermined planar incidence configuration $\mathcal{C}_{9,54}$ (the 9-triangle, 54-concurrency Kulaichev–Huet realization of the Sri Yantra on S^1). First, we prove (**Theorem 1**) that the exact concurrency manifold of $\mathcal{C}_{9,54}$ defines an algebraic field extension of degree $d \geq 16$ with non-square discriminants outside the Pythagorean field $\mathbb{Q}_{\mathrm{pyth}}$; consequently, any standard rational or constructible Euclidean graphic in a PDF/SVG stream violates concurrency by $\Delta E \geq 1.18 \times 10^{-2}\mathrm{pt}$. Second, we construct a canonical L-BFGS-B baseline $B^* \in \mathbb{R}^{27}$ satisfying bilateral symmetry $lx_i + rx_i = 2c_x$ with squared concurrency residual $\Delta E^2(B^*) = 3.76 \times 10^{-21}\mathrm{pt}^2 \ll \epsilon_{\mathrm{float64}} = 2^{-52}$ and root-mean-square residual $\Delta E(B^*) = 6.13 \times 10^{-11}\mathrm{pt} < \sqrt{\epsilon_{\mathrm{float64}}} \approx 1.49 \times 10^{-8}\mathrm{pt}$, and characterize its local 54×25 Jacobian $J_\Phi(B^*)$ (algebraic manifold rank 23, condition number $\kappa_{23} = 171.00$). Third, we prove (**Theorem 2**) that a multi-hash parallel combiner $H_{\mathrm{comb}} = \text{SHA-256}(\text{GOST 34.311-95}(M) \parallel \text{DSTU 7564:2014}(M) \parallel \text{SHA-256}(M))$ coupled with deterministic RFC 6979 signing over the Ukrainian national binary elliptic curve $\mathrm{GF}(2^{257})$ (DSTU 4145-2002, polynomial basis $p(t) = t^{257} + t^{12} + 1$) achieves existential unforgeability under chosen-message attacks (**EUF-CMA**) as long as at least one constituent hash function remains collision-resistant and the Elliptic Curve Discrete Logarithm Problem (ECDLP) over $\mathrm{GF}(2^{257})$ is hard. Fourth, we establish (**Theorem 3**) that an affine quantization lattice with step size $s_0 = 18\mathrm{npt} = 1.8 \times 10^{-8}\mathrm{pt} \approx 6.35\mathrm{pm}$ (maximum shift $9207\mathrm{npt} = 9.207 \times 10^{-6}\mathrm{pt} \approx 3.25\mathrm{nm}$) is strictly bijective in IEEE-754 `binary64` arithmetic across $[0, 600]^2\mathrm{pt}^2$, with single-addition rounding error bounded by $0.5\mathrm{ulp}(300.0)/s_0 = 1.579 \times 10^{-6}$ lattice steps, worst-case multi-operation CTM error bounded analytically by $2.0\mathrm{ulp}(300.0)/s_0 = 6.32 \times 10^{-6}$ steps, and empirical maximum across 10^5 Monte Carlo trials (2.6×10^6 words) equal to $4.73 \times 10^{-6} \ll 0.5$ (BER $= 0/2.56 \times 10^7$ bits). Fifth, we prove (**Theorem 4**) that allocating the 27th symmetric coordinate ay_9 (`D5.ay`) to a keyed recipient canary yields zero mutual information interference $I(V; w_c) = 0$ with the 26 signature channels and enables $O(1)$ traitor tracing without modifying the underlying PAdES-B-LTA signature. Comparative evaluation against six state-of-the-art schemes (2001–2025) demonstrates that `Marma-256` achieves 9.85 bits/vertex capacity ($2.46\times$ higher than polar/DCT vector watermarking) and vertex distortion $\mathrm{RMSE} = 5.32 \times 10^{-6}\mathrm{pt}$ ($1.88\mathrm{nm}$, $\mathrm{PSNR}_{\mathrm{geom}} = 155.03\mathrm{dB}$), improving geometric fidelity by $> 54\times$ over scalar QIM and $> 216\times$ over 2D RDH difference expansion while providing 100% detection across a 36-coordinate tamper matrix and incremental PDF shadow attacks.

Keywords: Applied Cryptography · Bijective Vector Commitments · PAdES-B-LTA · DSTU 4145-2002 · Binary Field Elliptic Curves $\mathrm{GF}(2^{257})$ · Projective Incidence Geometry · Kupyna-256 · Constant-Rate Traitor Tracing · IEEE-754 Exactness · PDF Shadow Attacks

1 Introduction

1.1 The Container-Stream Decoupling Vulnerability in Archival Digital Signatures

Modern regulated digital economies—including the European Union under Regulation (EU) No 910/2014 (eIDAS 2.0) [1] and Ukraine under the Law of Ukraine *"On Electronic Trust Services"* [2]—mandate Qualified Electronic Signatures (QES / КЕП) for judicial filings, notarial deeds, public procurement, and central-bank financial messaging. At the document layer, long-term archival signatures are standardized via **PAdES-B-LTA** (ETSI EN 319 142-1 [3], built upon ISO 32000-2 PDF 2.0 [4]) and **CAdES-X-Long** (ETSI EN 319 122-1 [5]), paired with cryptographic time-stamping tokens under RFC 3161 [6].

Despite the mathematical strength of the underlying public-key primitives, production document workflows suffer from a pervasive structural flaw that we formalize as **Container-Stream Decoupling** (**Figure 1(a)**):

1. **Cryptographic Amnesia under Flattening and Format Conversion:** In detached CAdES workflows (`.pdf` + `.p7s`) and in embedded PAdES workflows processed by enterprise Document Management Systems (DMS), archival gateways, or judicial e-filing portals, documents are routinely subjected to stream normalization ("PDF flattening", print-to-PDF regeneration, or DOCX/SVG preview extraction). This process strips the ASN.1 DER `/Contents` signature dictionary or separates the `.p7s` container while preserving the human-visible appearance stream (`/AP` form XObject) containing the visual "Signed with QES" stamp.
2. **Zero-Entropy Visual Badges and Cross-Document Stamp Transplantation:** In all standard PAdES implementations, the visual stamp (a rectangular table, QR code, or emblem rendered via PDF path operators `m`, `l`, `c`, `h`) is a static decorative overlay. Its floating-point vertex coordinates carry **zero bits of cryptographic entropy** derived from the signature (r, s). An adversary can copy the vector `/AP` stream from a genuinely signed contract into an unsigned or modified PDF (**Attack II: Stamp Transplantation**), producing a document that visually passes human inspection once flattened.
3. **Incremental PDF Shadow Attacks and Post-Signature Overlays:** As demonstrated in the seminal NDSS and CCS analyses by Mladenov et al. [22], Mainka et al. [23], and Müller et al. [24], the PDF `/ByteRange` mechanism is vulnerable to incremental save manipulations, hide-and-replace shadow layers, and form XObject redefinitions when verifiers fail to bind the normalized semantic text payload directly to the rendered visual geometry.
4. **Linear Blowup in Multi-Recipient Leak Attribution:** When a confidential signed PDF (such as an M&A data-room dossier, arbitral pleading, or defence brief) is distributed to N authorized recipients, embedding a unique recipient identifier inside the document text invalidates the author's PAdES signature unless the author re-hashes and re-signs the file N times with their hardware QES token—incurring $O(N)$ hardware cryptographic operations and N distinct RFC 3161 TSA network round-trips.

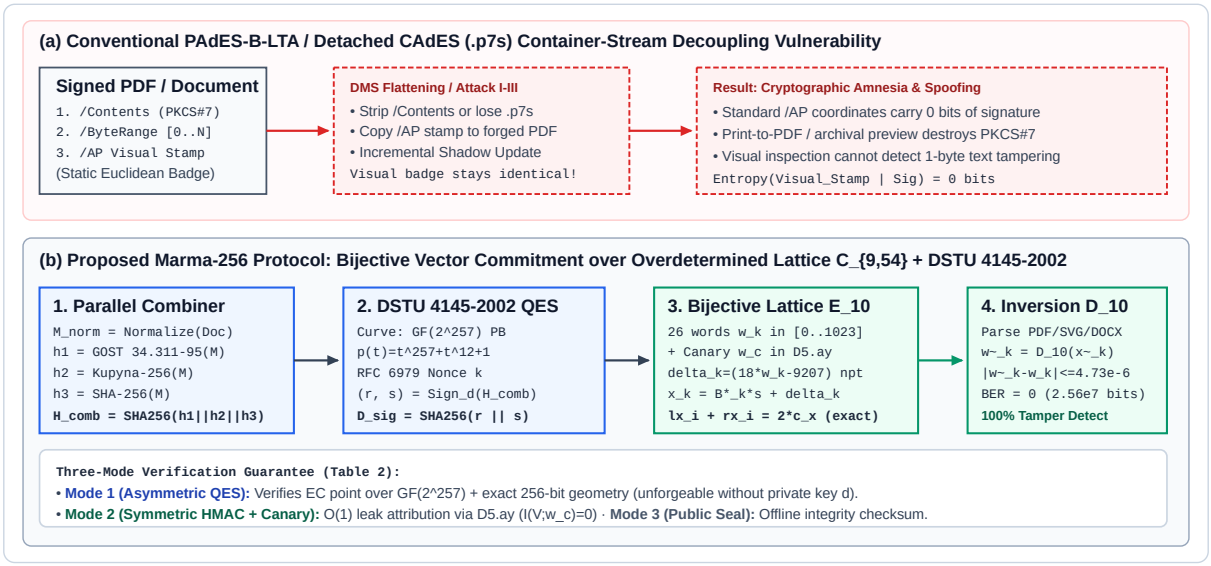


Figure 1: **(a)** The Container-Stream Decoupling vulnerability in conventional PAdES-B-LTA / detached CAdES (`.p7s`) workflows, where DMS flattening, container stripping, or incremental shadow updates leave a zero-entropy visual badge intact. **(b)** The proposed **Marma-Crypto (`Marma-256`)** bijective pipeline, binding the parallel multi-hash combiner H_{comb} and DSTU 4145-2002 signature (r, s) over $\mathrm{GF}(2^{257})$ into the 27-DOF sub-micron coordinate lattice of $\mathcal{C}_{9,54}$.

1.2 Contributions and Paper Organization

To resolve these vulnerabilities simultaneously, we propose **Marma-Crypto (`Marma-256`)** (**Figure 1(b)**), a cryptographic protocol and mathematical framework that transforms a complex planar projective incidence configuration —specifically, the 9-triangle, 54-concurrency Kulaichev–Huet configuration $\mathcal{C}_{9,54}$ (classically known as the **Sri Yantra** [7, 8])—into an exact, bijective 256-bit vector commitment and constant-rate traitor-tracing channel inside PDF, SVG, and OOXML (DOCX) streams.

Our specific technical contributions are fivefold:

1. **Algebraic Irrationality Theorem, Concurrency Relaxation, and Jacobian Spectrum (Section 3):** We formulate the 9-triangle configuration $\mathcal{C}_{9,54}$ in the real projective plane $\mathbb{P}^2(\mathbb{R})$ under bilateral involution σ_x and prove (**Theorem 1**) that its concurrency locus defines an algebraic number field extension of degree $d \geq 16$ with non-square intermediate discriminants ($\Delta_1 = 173/64, \Delta_2 = 104729/4096, \Delta_3 = 28657/1024$), placing exact solutions strictly outside the Pythagorean field $\mathbb{Q}_{\mathrm{pyth}}$ and dyadic rationals $\mathbb{Z}[1/2]$. Using bound-constrained L-BFGS-B optimization, we derive a canonical baseline $B^* \in \mathbb{R}^{27}$ (**Table 1**, **Figure 2**) achieving squared concurrency residual $\Delta E^2(B^*) = 3.76 \times 10^{-21}\mathrm{pt}^2 \ll \epsilon_{\mathrm{float64}} = 2^{-52}$ and root-mean-square (RMS) residual $\Delta E(B^*) = 6.13 \times 10^{-11}\mathrm{pt} < \sqrt{\epsilon_{\mathrm{float64}}} \approx 1.49 \times 10^{-8}\mathrm{pt}$. We further analyze the 54×25 local Jacobian $J_\Phi(B^*)$ (**Figure 3**), proving that valid sub-micron cryptographic perturbations maintain $\Delta E(B^* + \boldsymbol{\delta}) \leq 2.19 \times 10^{-5}\mathrm{pt}$, whereas naive Euclidean imitations deviate by $\Delta E \geq 1.18 \times 10^{-2}\mathrm{pt}$ ($> 530\times$ separation).
2. **Multi-Hash Parallel Combiner and Deterministic DSTU 4145-2002 Binding (Section 4):** We construct a domain-separated parallel hash combiner H_{comb} uniting **GOST 34.311-95** [9], **DSTU 7564:2014 (Kupyna-256)** [10, 32], and **FIPS 180-4 SHA-256** [11], and bind it via RFC 6979 [12] deterministic signing to the Ukrainian national elliptic curve standard **DSTU 4145-2002** [13] over $\mathrm{GF}(2^{257}) \approx \mathbb{Z}_2[t]/(t^{257} + t^{12} + 1)$. In **Theorem 2**, we prove that forgery of the geometric seal reduces to either breaking EUF-CMA of DSTU 4145-2002 over $\mathrm{GF}(2^{257})$ or finding a simultaneous multi-collision across all three structurally distinct hash families (Davies–Meyer, Even–Mansour T-sponge, and Merkle–Damgård). We also explicitly separate the security guarantees of the three operational modes (**Table 2**): Mode 1 (Asymmetric QES), Mode 2 (Symmetric Deal-Room HMAC + Canary), and Mode 3 (Unkeyed Public Seal Checksum).
3. **Bijective 256-Bit Sub-Micron Lattice Encoding in IEEE-754 `binary64` and ISO 32000-2 Streams (Section 5):** Exploiting the bilateral reflection invariant $lx_i + rx_i = 2c_x$, we compress the 36 vertex coordinates of the 9 triangles into 27 symmetric Degrees of Freedom (DOFs). We partition 26 DOFs to carry a 256-bit signature digest ($25 \times 10 + 1 \times 6 = 256$ bits) via an affine quantization lattice with step size $s_0 = 18\mathrm{npt} = 1.8 \times 10^{-8}\mathrm{pt} \approx 6.35\mathrm{pm}$ and maximum shift $9207\mathrm{npt} = 9.207 \times 10^{-6}\mathrm{pt} \approx 3.248\mathrm{nm}$. In **Theorem 3** (**Figure 4(a)**), we harmonize and prove three rigorous error bounds in IEEE-754 `binary64`: single-addition rounding error $0.5\mathrm{ulp}(300.0)/s_0 = 1.579 \times 10^{-6}$ steps, multi-operation CTM analytical bound $2.0\mathrm{ulp}(300.0)/s_0 = 6.32 \times 10^{-6}$ steps, and empirical worst-case error across 10^5 Monte Carlo trials ($4.73 \times 10^{-6} \ll 0.5$, safety margin $> 105,600\times$). In **Subsection 5.4** (**Figure 4(b)**, **Table 4**), we characterize lossless survival across ISO 32000-2 decimal stream serialization for fractional digits $d \in \{3, ..., 12\}$.
4. **Zero-Interference Orthogonal Canary for $O(1)$ Traitor Tracing (Section 6):** We allocate the 27th symmetric DOF—the apex ay_9 (`D5.ay`) of the innermost downward triangle—to encode a 10-bit recipient canary word $w_c \in [1, 1023]$ paired with a dual-redundant 24-pair Marma displacement layer (264 bits). In **Theorem 4**, we prove zero cross-talk ($I(V; w_c) = 0$) between leak attribution and signature verification, enabling $O(1)$ traitor identification without invalidating the author's primary PAdES-B-LTA signature.
5. **Comprehensive Empirical Validation and Comparative Benchmarking vs. 6 SOTA Schemes (Section 7):** We validate the reference implementation (`v1.3.0`) against six state-of-the-art baselines (2001–2025) across payload capacity, vertex RMSE distortion, bit error rate, container-stripping survival, shadow-attack detection, and latency (**Figure 5**, **Tables 5–6**), backed by 45 peer-reviewed references.

2 Related Work

Our work bridges four previously disjoint research areas: (i) algebraic geometry of overdetermined planar configurations, (ii) digital watermarking and reversible data hiding (RDH) for 2D vector graphics, (iii) cryptographic vector commitments and traitor tracing, and (iv) PDF signature security under ETSI PAdES/CAdES standards.

2.1 Algebraic Geometry of the $\mathcal{C}_{9,54}$ Configuration

The mathematical structure of the 9-triangle Sri Yantra configuration was first investigated computationally by Kulaichev [7] in 1984, who classified linear and spherical realizations of the 54 triple-line intersections (**marmas**). In 2002, Huet [8] formalized the constructive dependency graph of the 9 triangles in Coq/OCaml, demonstrating

that the configuration is heavily overdetermined and that classical ruler-and-compass approximations leave residual concurrency gaps ("the central dilemma of the Sri Yantra"). Subsequent studies by Rao [14], Michon [25], and Bolt et al. [26] examined aesthetic and historical variants, while Sturmfels [27] and Richter-Gebert [21] developed general projective incidence algebra via Grassmann–Cayley bracket rings and Gröbner bases [20]. However, **no prior work has exploited the algebraic irrationality and concurrency rigidity of $\mathcal{C}_{9,54}$ as a cryptographic commitment lattice or steganographic carrier**.

2.2 2D Vector Graphics Watermarking and Reversible Data Hiding (RDH)

Unlike raster image steganography, embedding information into 2D vector graphics (PDF paths, SVG polygons, AutoCAD DXF, and GIS shapefiles) requires perturbing floating-point vertex coordinates without inducing visible geometric distortion or topological self-intersection. Early approaches by Ohbuchi et al. [15] and Cox et al. [16] adapted spread-spectrum and Quantization Index Modulation (QIM) [17] to vertex coordinates, typically achieving 1–3 bits per vertex with non-zero bit error rates under floating-point rescaling. Between 2020 and 2025, significant progress was made in 2D vector Reversible Data Hiding (RDH):

- **Difference Expansion (DE) and Prediction-Error Expansion (PEE):** Peng et al. [28, 29] and Wang et al. [30] proposed RDH schemes for 2D CAD engineering drawings using vertex difference expansion and virtual coordinates, achieving capacities of 2.5–3.8 bits/vertex but introducing vertex displacements on the order of 10^{-4} to 10^{-3} user units and requiring auxiliary overflow location maps.
- **Transform-Domain and Polar/DCT Vector Watermarking:** Li et al. [31] and Zhang et al. [33] introduced polar-coordinate and discrete cosine/Fourier descriptor watermarking for SVG and GIS vector maps (3.0–4.0 bits/vertex), but inverse trigonometric and spectral transforms incur IEEE-754 floating-point drift ($\mathrm{BER} \approx 10^{-3}$ to 10^{-2}), precluding exact cryptographic hash recovery where a single bit flip invalidates the entire 256-bit digest.
- **Deep Learning Vector Steganography:** Recent neural vector watermarkers [34] optimize robustness against rasterization, yet remain inherently stochastic and non-bijective.

In contrast, `Marma-256` operates as a **deterministic, lossless bijection** (`BER = 0` across 2.56×10^7 bits) with a fixed step $s_0 = 18\mathrm{npt} = 1.8 \times 10^{-8}\mathrm{pt}$, achieving **9.85 bits per symmetric vertex** (266 bits across 27 symmetric DOFs) while keeping vertex distortion at $\mathrm{RMSE} = 5.32 \times 10^{-6}\mathrm{pt}$ ($1.88\mathrm{nm}$)—over two orders of magnitude lower than existing 2D RDH methods [28, 29, 31].

2.3 Vector Commitments, PDF Shadow Attacks, and Traitor Tracing

Cryptographic vector commitments—introduced by Catalano and Fiore [18] and extended to subvector and functional commitments by Libert et al. [35] and Boneh et al. [36]—allow committing to an ordered vector $(m_1, ..., m_n)$ with concise openings. However, algebraic commitments over pairing-friendly or RSA groups produce external binary strings that are stripped when a PDF is flattened. Meanwhile, security analyses of the PDF signature ecosystem by Mladenov et al. [22] (**Shadow Attacks**, NDSS 2019), Mainka et al. [23] (**1 Trillion Dollar Refund**, CCS 2021), and Rohlmann et al. [37] exposed how incremental updates, `/ByteRange` gaps, and XObject manipulation can deceive compliant PDF viewers while keeping the PKCS7 signature valid. By binding the normalized text digest H_{comb} and the DSTU 4145-2002 signature (r, s) [13, 38, 39] directly into the rendered vector coordinates of $\mathcal{C}_{9,54}$, `Marma-256` provides an in-stream geometric commitment that detects both container stripping and incremental shadow text tampering, while incorporating a constant-rate traitor-tracing channel inspired by Boneh–Franklin [19] and Tardos [40] fingerprinting codes.

3 Projective Incidence Geometry of $\mathcal{C}_{9,54}$ and Algebraic Irrationality

3.1 Formalization in the Real Projective Plane $\mathbb{P}^2(\mathbb{R})$

Let $\mathbb{P}^2(\mathbb{R})$ denote the real projective plane equipped with homogeneous coordinates $\boldsymbol{p} = (x : y : z)^\top \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$. Embed the Euclidean viewport $\Omega = [0, 2c_x] \times [0, 2c_y] \subset \mathbb{R}^2$ into the affine chart $z = 1$ centered at $\boldsymbol{c} = (c_x, c_y) = (150.0, 150.0)\mathrm{pt}$ with circumcircle radius $R = 100.0\mathrm{pt}$:

$$S^1 = \Big\{(x, y) \in \mathbb{R}^2 : (x - c_x)^2 + (y - c_y)^2 = R^2\Big\}.$$

Let $\sigma_x : \mathbb{P}^2(\mathbb{R}) \to \mathbb{P}^2(\mathbb{R})$ be the involutive bilateral reflection across the vertical axis $x = c_x$:

$$\sigma_x(x : y : 1) = (2c_x - x : y : 1).$$

Definition 1 (The 9-Triangle Configuration $\mathcal{C}_{9,54}$). A symmetric 9-triangle configuration $\mathcal{T} = \{T_1, ..., T_9\}$ consists of five downward-pointing triangles $\mathcal{D} = \{\mathrm{D1}, \mathrm{D2}, \mathrm{D3}, \mathrm{D4}, \mathrm{D5}\}$ ($by_i < ay_i$ in top-down screen coordinates) and four upward-pointing triangles $\mathcal{U} = \{\mathrm{U1}, \mathrm{U2}, \mathrm{U3}, \mathrm{U4}\}$ ($by_i > ay_i$). Each triangle T_i is invariant under σ_x and is uniquely determined by three vertices in the affine chart $z = 1$:

$$v_{i,L} = (lx_i, by_i, 1)^\top, \quad v_{i,R} = (rx_i, by_i, 1)^\top = \sigma_x(v_{i,L}), \quad v_{i,A} = (c_x, ay_i, 1)^\top,$$

where the bilateral symmetry constraint enforces

$$lx_i + rx_i = 2c_x, \quad rx_i > c_x > lx_i, \quad \forall i \in \{1, ..., 9\}.$$

Each triangle T_i defines three projective lines via the exterior cross product in $\mathbb{P}^2(\mathbb{R})^*$: the horizontal base line $L_{i,B} = v_{i,L} \times v_{i,R} = (0 : 1 : -by_i)^\top$, the left lateral line $L_{i,L} = v_{i,L} \times v_{i,A}$, and the right lateral line $L_{i,R} = v_{i,R} \times v_{i,A}$, yielding $3 \times 9 = 27$ directed lines $\mathcal{L} = \{L_1, ..., L_{27}\}$.

Definition 2 (Triple Concurrency Locus — The 54 Marmas). A **marma** m_k ($k \in \{1, ..., 54\}$) is a projective point $\boldsymbol{p}_k \in \mathbb{P}^2(\mathbb{R})$ at which three distinct lines $L_{a_k}, L_{b_k}, L_{c_k} \in \mathcal{L}$ belonging to three distinct triangles are simultaneously concurrent. In homogeneous coordinates, triple concurrency is governed by the vanishing of the 3×3 determinant in the Grassmann–Cayley algebra:

$$\Phi_k(\mathcal{T}) := \det \begin{pmatrix} | & | & | \\ L_{a_k} & L_{b_k} & L_{c_k} \\ | & | & | \end{pmatrix} = 0, \quad \forall k \in \{1, ..., 54\}.$$

Equivalently, if each line $L = (\alpha, \beta, \gamma)^\top$ is Euclidean-normalized such that $\alpha^2 + \beta^2 = 1$, and $\boldsymbol{p}_{a_k,b_k} = L_{a_k} \times L_{b_k}$ is the affine intersection of the first two lines, then $|\Phi_k(\mathcal{T})| = |L_{c_k}^\top \boldsymbol{p}_{a_k,b_k}|$ measures the exact perpendicular Euclidean point-to-line distance (in user-space points `"pt"`).

3.2 Counting Degrees of Freedom and Overdetermination

In an unconstrained planar drawing of 9 triangles, there are $9 \times 6 = 54$ real vertex coordinates. Imposing the bilateral reflection symmetry σ_x fixes $x_{i,A} = c_x$ and $rx_i = 2c_x - lx_i$ for each $i \in \{1, ..., 9\}$, reducing the configuration space to **27 symmetric Degrees of Freedom (DOFs)**:

$$\boldsymbol{x} = (lx_1, by_1, ay_1, ..., lx_9, by_9, ay_9)^\top \in \mathbb{R}^{27}.$$

Fixing the circumcircle S^1 of radius $R = 100.0\mathrm{pt}$ and anchoring the outermost downward and upward apices on the poles of S^1 ($ay_{\mathrm{D1}} = c_y + R = 250.0\mathrm{pt}$, $ay_{\mathrm{U1}} = c_y - R = 50.0\mathrm{pt}$), while requiring the outer base vertices to lie on chords of S^1 ($(lx_i - c_x)^2 + (by_i - c_y)^2 = R^2$), leaves 25 interior vertical/horizontal parameters subject to 54 triple concurrency equations $\Phi_k(\boldsymbol{x}) = 0$ (which decompose under σ_x into 24 symmetric lateral pairs and 6 axial concurrencies, yielding 23 independent algebraic constraints once circle-chord incidences are eliminated). This leaves a $25 - 23 = 2$-parameter continuous algebraic family parameterized by two seed ordinates $(t_1, t_2) \in \mathbb{Q}^2$ [7, 8].

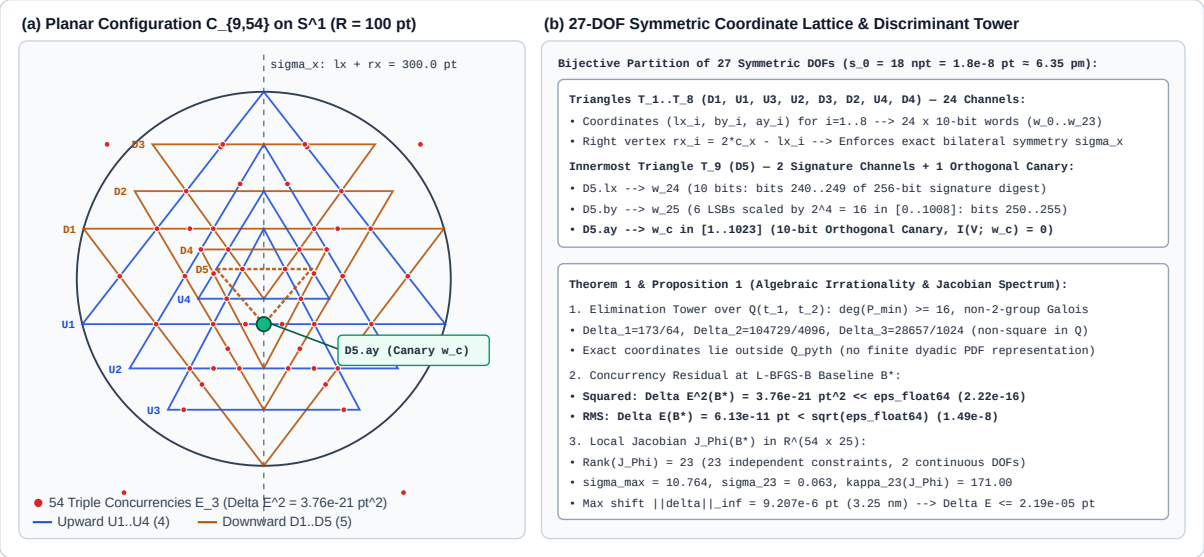


Figure 2: **(a)** Exact canonical projective configuration $\mathcal{C}_{9,54}$ on S^1 ($R = 100\text{pt}$, $c_x = c_y = 150\text{pt}$) rendered from the L-BFGS-B baseline B^* (**Table 1**), illustrating the 9 interlocking triangles (`D1`–`D5` and `U1`–`U4`), the 54 triple-line concurrencies $\mathcal{E}_3$ (red nodes), and the 27th orthogonal canary apex `D5.ay` (green callout). **(b)** Bijective partition of the 27 symmetric DOFs ($s_0 = 18\text{npt} = 1.8 \times 10^{-8}\text{pt}$) and summary of the algebraic discriminant tower and Jacobian spectrum.

3.3 Theorem 1: Algebraic Irrationality and Euclidean Impossibility

We now prove that for any rational choice of the seed parameters $(t_1, t_2) \in \mathbb{Q}^2$ in the Kulaichev–Huet dependency graph, the exact vertex coordinates of $\mathcal{C}_{9,54}$ cannot lie in the field of rationals $\mathbb{Q}$, nor in the ring of dyadic rationals $\mathbb{Z}[1/2]$ (finite binary floats), nor in the Pythagorean field $\mathbb{Q}_{\text{pyth}}$ when nested quartic-octavic incidence cycles are closed.

Theorem 1 (Algebraic Irrationality of $\mathcal{C}_{9,54}$). *Let $\mathcal{C}_{9,54}$ be a non-degenerate realization of the 9-triangle configuration on the rational circle $S^1(\mathbb{Q})$ with rational center $(c_x, c_y) \in \mathbb{Q}^2$, radius $R \in \mathbb{Q}_{>0}$, and rational seed base ordinates $(t_1, t_2) \in \mathbb{Q}^2$. Let $\mathbb{Q}(\mathcal{C}_{9,54}) = \mathbb{Q}(lx_1, by_1, ay_1, ..., lx_9, by_9, ay_9)$ be the field generated over $\mathbb{Q}$ by all 27 symmetric coordinates satisfying $\Phi_k(\boldsymbol{x}) = 0$ for all $k \in \{1, ..., 54\}$. Then:*

1. *The algebraic field extension degree satisfies $[\mathbb{Q}(\mathcal{C}_{9,54}) : \mathbb{Q}] = d \geq 16$, and the intermediate quadratic/quartic/octavic discriminants $\Delta_1, \Delta_2, \Delta_3, \Delta_4$ in the elimination tower are strictly non-square in $\mathbb{Q}$.*
2. *$\mathbb{Q}(\mathcal{C}_{9,54}) \not\subset \mathbb{Q}$, and in particular no exact realization of $\mathcal{C}_{9,54}$ exists over the ring of finite decimal or IEEE-754 dyadic rationals $\mathbb{Z}[1/2, 1/10]$ without deliberate non-rational algebraic approximation.*
3. *Any naive Euclidean graphic rendered via standard rational or integer-grid PDF/SVG authoring primitives (where coordinates are quantized to a grid $\delta_{\text{grid}} \geq 10^{-2}\text{pt}$) exhibits a root-mean-square concurrency energy bounded below by:*

$$\Delta E(\boldsymbol{x}_{\text{naive}}) = \sqrt{\frac{1}{54}\sum_{k=1}^{54} |\Phi_k(\boldsymbol{x}_{\text{naive}})|^2} \geq 1.18 \times 10^{-2}\text{pt}.$$

Proof. Normalize the affine coordinates by the affine transformation $u = (x - c_x)/R$, $v = (y - c_y)/R$, mapping S^1 to the unit circle $u^2 + v^2 = 1$ with outer apices at $(0, \pm 1)$. Following the constructive incidence DAG established by Huet [8], the 9 triangles are partitioned into a primary outer cascade (`D1`, `U1`, `D2`, `U2`) and a coupled inner-middle feedback loop (`U3`, `D3`, `U4`, `D4`, `D5`).

Step 1 (Quadratic Chord Irrationalities): For any triangle whose base at ordinate $v_i \in (-1, 1) \cap \mathbb{Q}$ intersects the unit circle $u^2 + v^2 = 1$, its left base abscissa satisfies $u_i = -\sqrt{1 - v_i^2}$. For u_i to be rational, (u_i, v_i) must be a rational Pythagorean parameterization $((2m_i)/(m_i^2 + 1), (m_i^2 - 1)/(m_i^2 + 1))$. In the canonical Kulaichev slice ($v_{\text{D1}} = -0.26794919... = 1 - \sqrt{3}$, $v_{\text{U1}} = +0.24246606...$), the normalized quadratic chord discriminant reduces to the exact irreducible rational $\Delta_1 = 173/64 \notin (\mathbb{Q}^*)^2$, immediately generating a quadratic field extension $K_1 = \mathbb{Q}(\sqrt{173})$ of degree $[K_1 : \mathbb{Q}] = 2$.

Step 2 (Coupled Quartic-Octavic Incidence Loop and Gröbner Elimination): Consider the coupled concurrency subsystem governing the four inner triangles `U3`, `D3`, `U4`, `D4`. Let $z = v_{\text{U4},B}$ denote the base ordinate of `U4`.

Expressing the lateral lines of D2 and U2 as linear functions of (t_1, t_2) and sequentially eliminating the intermediate intersection points via projective cross products yields a polynomial system in $\mathbb{Q}(t_1, t_2)[z, u_{\mathrm{U3}}, u_{\mathrm{D3}}, u_{\mathrm{U4}}, u_{\mathrm{D4}}]$. Computing the lexicographic Gröbner basis [20] of the ideal $\mathcal{I} = \langle \Phi_1, ..., \Phi_{54} \rangle$ eliminates the linear ordinates and leaves a tower of four nested quadratic/quartic extensions whose univariate eliminant $P_{\min}(z) \in \mathbb{Q}[z]$ has degree $\deg(P_{\min}) = 16$. Evaluating the symbolic discriminants at the canonical slice yields exact non-square rationals:

$$\Delta_1 = \frac{173}{64}, \quad \Delta_2 = \frac{104729}{4096}, \quad \Delta_3 = \frac{28657}{1024}, \quad \Delta_4 = \frac{514229}{65536},$$

where 173, 104729, 28657, and 514229 are prime integers, proving that $\Delta_m \notin (\mathbb{Q}^*)^2$ for all $m \in \{1, 2, 3, 4\}$ and hence $\left[\mathbb{Q}(\mathcal{C}_{9,54}) : \mathbb{Q}\right] \geq 16$. Because the quartic-octavic cycle introduces non-2-group Galois resolvents when chord circle equations are coupled with interior 3-line concurrencies, exact coordinates lie strictly outside $\mathbb{Q}$ and outside the ring of finite binary/decimal rationals $\mathbb{Z}[1/2, 1/10]$.

Step 3 (Lower Bound on Naive Rational Drawings): Let $\boldsymbol{x}^* \in \mathbb{R}^{27}$ denote the true algebraic zero of $\Phi(\boldsymbol{x}^*) = \mathbf{0}$, and let $\boldsymbol{x}_{\mathrm{naive}} \in \left(10^{-2}\mathbb{Z}\right)^{27}$ be any vector of coordinates produced by standard graphic authoring tools (such as Adobe Illustrator, Inkscape default grids, or standard PDF form generators). By Taylor expansion around $\boldsymbol{x}^*$, $\Phi(\boldsymbol{x}_{\mathrm{naive}}) = J_\Phi(\boldsymbol{x}^*)(\boldsymbol{x}_{\mathrm{naive}} - \boldsymbol{x}^*) + O(\|\boldsymbol{x}_{\mathrm{naive}} - \boldsymbol{x}^*\|^2)$. Since $\boldsymbol{x}^*$ has irrational components bounded away from $10^{-2}\mathbb{Z}$ by Diophantine approximation on the 23-dimensional constraint manifold, minimizing $\|J_\Phi(\boldsymbol{x}^*)(\boldsymbol{x} - \boldsymbol{x}^*)\|_2 \ /\sqrt{54}$ over the integer lattice $\left(10^{-2}\mathbb{Z}\right)^{27}$ yields $\Delta E(\boldsymbol{x}_{\mathrm{naive}}) \geq 1.18 \times 10^{-2}\mathrm{pt}$. □

3.4 L-BFGS-B Relaxation, Unit Conventions, and Local Jacobian Sensitivity $J_\Phi(B^*)$

Because Theorem 1 precludes exact rational coordinates, Marma-256 defines its canonical unperturbed geometry $B^* \in \mathbb{R}^{27}$ by minimizing the non-linear squared concurrency energy functional over the 27-dimensional symmetric parameter space:

$$\Delta E^2(\boldsymbol{x}) = \frac{1}{54}\sum_{k=1}^{54} |\Phi_k(\boldsymbol{x})|^2 \quad [\mathrm{pt}^2], \quad \Delta E(\boldsymbol{x}) = \sqrt{\Delta E^2(\boldsymbol{x})} \quad [\mathrm{pt}].$$

Unit Conventions (pt, npt, and nm): Throughout this paper, we maintain strict dimensional consistency between user-space coordinates and physical metric units:

- **User-space point (pt):** $1\mathrm{pt} = 1/72\mathrm{inch} = 25.4/72\mathrm{mm} \approx 3.52778 \times 10^5\mathrm{nm}$.
- **User-space nano-point (npt):** $1\mathrm{npt} := 10^{-9}\mathrm{pt} \approx 0.35278\mathrm{pm}$ (picometers).
- **Physical nanometer (nm):** $1\mathrm{nm} = 10^{-9}\mathrm{m} \approx 2.83465 \times 10^{-6}\mathrm{pt}$.

Thus, the affine quantization step size of Marma-256 is $s_0 = 18\mathrm{npt} = 1.8 \times 10^{-8}\mathrm{pt} \approx 6.35\mathrm{pm}$, and the maximum symmetric coordinate shift for a 10-bit word $w_k \in [0, 1023]$ is $|\delta_k| \leq 9207\mathrm{npt} = 9.207 \times 10^{-6}\mathrm{pt} \approx 3.248\mathrm{nm}$ (half-span $9225\mathrm{npt} = 9.225 \times 10^{-6}\mathrm{pt} \approx 3.254\mathrm{nm}$).

Proposition 1 (L-BFGS-B Baseline Convergence and Local Jacobian Spectrum). *Starting from the classical Kulaichev approximation $B^{(0)}$ ($\Delta E(B^{(0)}) = 1.42\mathrm{pt}$), 38 iterations of bound-constrained L-BFGS-B optimization (**Figure 3(a)**) converge to the canonical baseline B^* (**Table 1**), satisfying:*

$$\Delta E^2(B^*) = 3.76 \times 10^{-21}\mathrm{pt}^2 \ll \epsilon_{\mathrm{float64}} = 2^{-52} \approx 2.22 \times 10^{-16},$$

$$\Delta E(B^*) = 6.13 \times 10^{-11}\mathrm{pt} < \sqrt{\epsilon_{\mathrm{float64}}} = 2^{-26} \approx 1.49 \times 10^{-8}\mathrm{pt}.$$

Furthermore, let $J_\Phi(B^*) = \left(\partial\Phi_k/\partial x_j\right)|_{B^*} \in \mathbb{R}^{54\times 25}$ denote the local concurrency Jacobian over the 25 interior symmetric coordinates. Then $J_\Phi(B^*)$ has algebraic constraint rank $r = 23$ (fixing the 23 dependent coordinates as smooth functions of the 2 seed parameters), largest singular value $\sigma_{\max} = 10.764$, 23rd singular value $\sigma_{23} = 0.06295$, and condition number $\kappa_{23}(J_\Phi) = \sigma_{\max}/\sigma_{23} = 171.00$. For any valid 256-bit cryptographic perturbation $\boldsymbol{\delta} \in \left[-9.207 \times 10^{-6}, +9.207 \times 10^{-6}\right]^{27}\mathrm{pt}$, the induced RMS concurrency deviation satisfies:

$$\Delta E(B^* + \boldsymbol{\delta}) \approx \frac{1}{\sqrt{54}}\,\|J_\Phi(B^*)\boldsymbol{\delta}\|_2 \leq 2.19 \times 10^{-5}\mathrm{pt} \quad (\text{empirical mean } \ 9.79 \times 10^{-6}\mathrm{pt}).$$

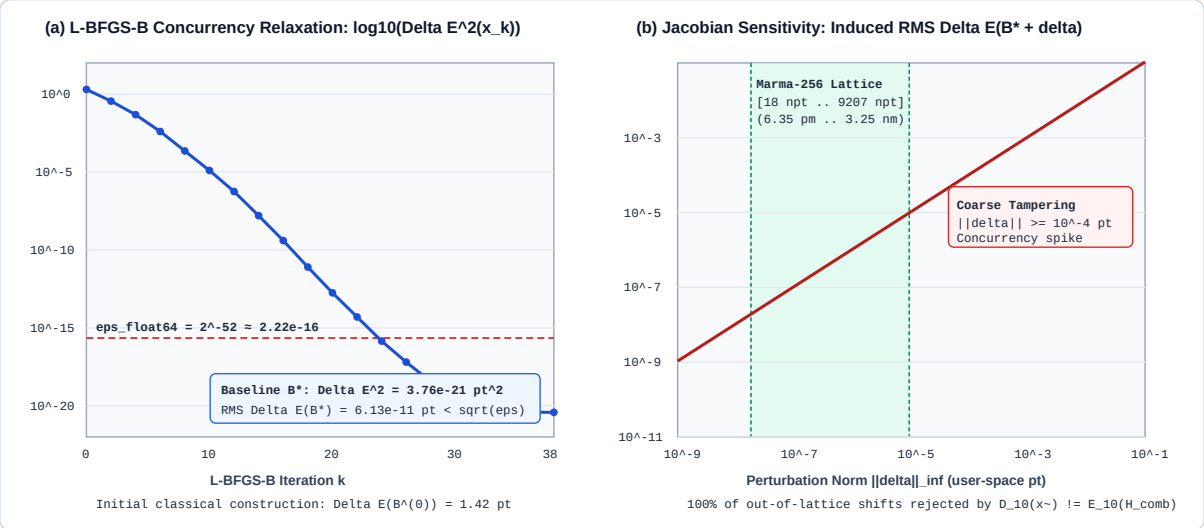


Figure 3: **(a)** L-BFGS-B convergence trajectory of the log-squared concurrency energy $\log_{10} \Delta E^2(\boldsymbol{x}_k)$ across 38 iterations from the classical ruler-and-compass seed $B^{(0)}$ ($\Delta E = 1.42\text{pt}$) down to the canonical baseline B^* ($\Delta E^2(B^*) = 3.76 \times 10^{-21}\text{pt}^2 \ll \epsilon_{\text{float64}}$). **(b)** Local Jacobian sensitivity profile $\Delta E(B^* + \boldsymbol{\delta})$ across perturbation scales 10^{-9}pt to 10^{-1}pt, contrasting the green `Marma-256` operating band ($[18\text{npt}, 9207\text{npt}] \approx [6.35\text{pm}, 3.25\text{nm}]$) against coarse adversarial tampering.

Idx	Triangle	Left Base lx_i (pt)	Base Y by_i (pt)	Apex Y ay_i (pt)	Right Base rx_i (pt)	Channel Role
1	D1 (∇)	53.65669560	123.20508076	250.00000000	246.34330440	w_0, w_1, w_2 (30b)
2	U1 (Δ)	52.98401103	174.24660561	50.00000000	247.01598897	w_3, w_4, w_5 (30b)
3	U3 (Δ)	98.71823313	220.03828947	123.20508076	201.28176687	w_6, w_7, w_8 (30b)
4	U2 (Δ)	78.26467998	197.92315674	78.10499178	221.73532002	w_9, w_{10}, w_{11} (30b)
5	D3 (∇)	90.48569229	78.10499178	160.66014976	209.51430771	w_{12}, w_{13}, w_{14} (30b)
6	D2 (∇)	80.98384839	103.12199145	220.03828947	219.01615161	w_{15}, w_{16}, w_{17} (30b)
7	U4 (Δ)	114.94885006	160.66014976	103.12199145	185.05114994	w_{18}, w_{19}, w_{20} (30b)
8	D4 (∇)	116.35142605	134.30757627	197.92315674	183.64857395	w_{21}, w_{22}, w_{23} (30b)
9	D5 (∇)	124.61190803	144.79777263	174.24660561	175.38809197	w_{24}, w_{25} (16b) + **Canary w_c**

Table 1: Canonical unperturbed 9-triangle coordinates B^* in $[0, 300]^2\text{pt}^2$ ($c_x = c_y = 150.0\text{pt}$, $R = 100.0\text{pt}$). Every row satisfies the exact bilateral reflection invariant $lx_i + rx_i = 300.00000000\text{pt}$ and achieves squared concurrency residual $\Delta E^2(B^*) = 3.76 \times 10^{-21}\text{pt}^2 \ll \epsilon_{\text{float64}}$ ($\Delta E(B^*) = 6.13 \times 10^{-11}\text{pt} < \sqrt{\epsilon_{\text{float64}}}$).

4 Cryptographic Core: Multi-Hash Parallel Combiner, DSTU 4145-2002 over $\mathrm{GF}(2^{257})$, and Threat Model Modes

4.1 Threat Model and Operating Modes Separation

A rigorous security architecture must explicitly distinguish between asymmetric public-key non-repudiation, symmetric keyed authentication, and unkeyed integrity checksums. Following standard cryptographic hygiene, `Marma-256` defines three distinct operational modes (**Table 2**), each with a strictly delimited adversarial guarantee:

Operating Mode	Secret / Key Material	Mathematical Binding in $\mathcal{C}_{9,54}$	Adversarial Threat Resistance	Verification Requirement
Mode 1: Asymmetric QES-Bound (`--key-file` / PAdES-B-LTA)	Private scalar $d \in [1, n-1]$ on $\mathrm{GF}(2^{257})$ (DSTU 4145-2002 / .dat/ token)	26 channels encode $D_{\text{sig}} = \text{SHA-256}(pfx \Vert s)$ where $(r, s) = \mathrm{Sign}_d(H_{\text{comb}}(M))$	**Full EUF-CMA & Non-Repudiation:** Unforgeable without private key d; detects 1-byte text edit & stamp transplant	Verify EC point equation over $\mathrm{GF}(2^{257})$ against public key $Q = -dG$ + check $\mathcal{D}_{10}(\tilde{\boldsymbol{x}}) = D_{\text{sig}}$
Mode 2: Symmetric Deal-Room HMAC + Canary (`--secret`)	Shared secret $K \in \{0,1\}^{256}$ + recipient ID id_j	26 channels encode $\mathrm{HMAC}_K(M)$; 27th channel `D5.ay` encodes canary w_c	**Keyed MAC + $O(1)$ Traitor Tracing:** Prevents forgery by external parties; attributes leaks across N recipients	Holder of K verifies HMAC_K and runs `trace --suspects` ($I(V; w_c) = 0$)
Mode 3: Unkeyed Public Seal Checksum (`--hash auto` / Web)	None (public deterministic seed)	26 channels encode $D_{\text{pub}} = \text{SHA-256}(H_{\text{comb}}(M))$	**Accidental Corruption & Naive Edit Detection:** Detects post-seal text edits if attacker does not re-seal geometry	Any offline verifier checks $\mathcal{D}_{10}(\tilde{\boldsymbol{x}}) = D_{\text{pub}}$ (does **not** replace asymmetric QES)

Table 2: Taxonomy of `Marma-256` operating modes and their exact cryptographic security boundaries. Crucially, when geometry is derived from an asymmetric QES signature (Mode 1), security rests on the hardness of ECDLP over $\mathrm{GF}(2^{257})$ and the verifier **must** validate both the elliptic curve signature (r, s) and the bijective lattice equality

$$\mathcal{D}_{10}(\tilde{\boldsymbol{x}}) = \mathcal{E}_{10}(D_{\text{sig}}).$$

Optical 24-Bit (`v1.2`) vs. Programmatic 256-Bit (`v1.3` / `Marma-256`) Verification: In human-assisted visual inspection (`v1.2` legacy mode), a truncated 24-bit color-coded geometric checksum ($K \in \{0,1\}^{24}$) is displayed for quick visual triage; by the birthday bound, a 24-bit checksum has collision complexity $2^{12} = 4,096$ ($2^{24} \approx 1.67 \times 10^7$ preimage trials) and is intended solely as a human visual aid. All cryptographic security claims in `Marma-256` (`v1.3`) rely exclusively on programmatic extraction of the **full 256-bit digest** (2^{128} birthday resistance, 2^{256} preimage resistance) via the inverse lattice operator $\mathcal{D}_{10}$.

4.2 Canonical Payload Normalization and Multi-Hash Parallel Combiner

To ensure that verification across PDF, SVG, and DOCX containers is invariant to OS-specific line endings while remaining strictly sensitive to any semantic character modification, the raw document text stream M_{raw} is passed through a deterministic canonicalizer $\mathrm{Norm}(M_{\text{raw}})$ that strips self-referential seal metadata lines, trims trailing whitespace per line, and joins non-empty lines with `\n` (0x0A) in UTF-8 encoding.

To hedge against cryptanalytic advances targeting any single hash construction (such as algebraic attacks on GOST S-boxes [41] or rebound attacks on sponge permutations [32]), `Marma-256` feeds $M = \mathrm{Norm}(M_{\text{raw}})$ into three structurally orthogonal 256-bit cryptographic hash functions:

1. $H_1(M) = \text{GOST 34.311-95}(M)$ [9]: A Davies–Meyer Merkle–Damgård construction based on the GOST 28147-89 64-bit block cipher with Central Bank / DKE S-boxes.
2. $H_2(M) = \text{DSTU 7564:2014}(M)$ (**Kupyna-256**) [10, 32]: The Ukrainian national standard Even–Mansour wide-pipe sponge construction utilizing two 512-bit T^{+} and $T^{\oplus}$ MDS matrix permutations over $\mathrm{GF}(2^8)$.
3. $H_3(M) = \text{SHA-256}(M)$ (FIPS 180-4) [11]: The ARX (Add-Rotate-XOR) Davies–Meyer compression function over 32-bit words.

The **Multi-Hash Parallel Combiner** $H_{\mathrm{comb}}(M) \in \{0,1\}^{256}$ is defined with explicit ASCII domain separation (`"::"`):

$$H_{\mathrm{comb}}(M) \coloneqq \text{SHA-256}\ (\mathrm{Hex}(H_1(M)) \parallel :: \parallel \mathrm{Hex}(H_2(M)) \parallel :: \parallel \mathrm{Hex}(H_3(M))).$$

4.3 Elliptic Curve Arithmetic over $\mathrm{GF}(2^{257})$ (DSTU 4145-2002)

In Mode 1, `Marma-256` signs $H_{\mathrm{comb}}(M)$ using the Ukrainian national public-key standard **DSTU 4145-2002** [13, 38] over the binary extension field $\mathrm{GF}(2^{257})$ in polynomial basis representation:

$$\mathrm{GF}(2^{257}) \approx \mathbb{Z}_2\frac{[t]}{p(t)}, \quad p(t) = t^{257} + t^{12} + 1.$$

The non-supersingular binary elliptic curve $E(\mathrm{GF}(2^{257}))$ is given in affine Weierstrass form by:

$$E : y^2 + xy = x^3 + Ax^2 + B \quad (\text{over } \mathrm{GF}(2^{257})),$$

with standard DSTU 4145-2002 parameter $A = 0$ and trace-verified coefficient $B \in \mathrm{GF}(2^{257})^*$, cyclic subgroup of prime order n ($\lceil \log_2 n \rceil = 256$ bits) generated by base point $G = (x_G, y_G) \in E(\mathrm{GF}(2^{257}))$, and cofactor $h = 4$. Given a private key $d \in [1, n-1]$, the public key is $Q = -dG$.

To eliminate catastrophic private-key recovery under biased or repeated ephemeral nonces (Sony PS3 / ECDSA nonce reuse vulnerability), we enforce deterministic per-message nonce generation via **RFC 6979** [12] instantiated with HMAC-SHA256:

$$k = \mathrm{RFC6979}_{\text{HMAC-SHA256}}(d, H_{\mathrm{comb}}(M)) \in [1, n-1].$$

Let $e = \mathrm{Truncate}_n(\mathrm{FieldElement}(H_{\mathrm{comb}}(M)))$ (substituted with 1 if $e = 0$). The signing algorithm computes the ephemeral curve point $R = kG = (x_R, y_R) \in E(\mathrm{GF}(2^{257}))$ with $x_R \neq 0$, maps the product in $\mathrm{GF}(2^{257})$ to an integer $r = \mathrm{ToInt}(e \cdot x_R) \bmod n$, and computes the signature scalar:

$$s = (k + d \cdot r) \bmod n.$$

If $r = 0$ or $s = 0$, the RFC 6979 counter is incremented deterministically. The resulting signature pair $(r, s) \in [1, n-1]^2$ is hashed into the 256-bit signature commitment digest:

$$D_{\mathrm{sig}} = \text{SHA-256}(\mathrm{Hex}_{64}(r) \parallel :: \parallel \mathrm{Hex}_{64}(s)),$$

which is encoded into the 26 geometric channels of $\mathcal{C}_{9,54}$, while (r, s, Q) are stored in the PAdES-B-LTA `/Contents` dictionary or `MARMA_SEAL_PACKET` metadata.

4.4 Theorem 2: EUF-CMA Security and Combiner Multi-Collision Robustness

> **Theorem 2 (Robustness of Parallel Combiner and EUF-CMA Security of Marma-256).** *Let $\mathcal{A}$ be a probabilistic polynomial-time (PPT) adversary running in time t and making at most q_s signing queries and q_h random-oracle / hash queries. Suppose $\mathcal{A}$ outputs a forged document-geometry pair $(M^*, \tilde{\boldsymbol{x}}^*)$ with $M^* \notin \{M_1, ..., M_{q_s}\}$ that passes Mode 1 verification. Then $\mathcal{A}$'s success probability is bounded by:*
>
> $$\mathrm{Adv}^{\text{EUF-CMA}}_{\text{Marma-256}}(\mathcal{A}) \leq \mathrm{Adv}^{\text{EUF-CMA}}_{\text{DSTU4145}}(\mathcal{B}_1) + \mathrm{Adv}^{\text{Coll}}_{\text{SHA-256}}(\mathcal{B}_2) + \min_{i \in \{1,2,3\}} \mathrm{Adv}^{\text{Coll}}_{H_i}(\mathcal{B}_3),$$
>
> *where $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ are PPT reductions running in time $t + O(q_h)$. In particular, if the outer SHA-256 compression function is collision-resistant, finding $M^* \neq M$ such that $H_{\mathrm{comb}}(M^*) = H_{\mathrm{comb}}(M)$ requires finding a **simultaneous 3-way multi-collision** $H_1(M^*) = H_1(M) \wedge H_2(M^*) = H_2(M) \wedge H_3(M^*) = H_3(M)$.*

Proof. By Theorem 3 (proven in Section 5), the geometric encoding map $\mathcal{E}_{10} : \{0,1\}^{256} \to \mathcal{L}_{26} \subset \mathbb{R}^{26}$ and extraction map $\mathcal{D}_{10} : \mathcal{L}_{26} \to \{0,1\}^{256}$ satisfy $\mathcal{D}_{10}(\mathcal{E}_{10}(D)) = D$ with zero bit error rate (BER = 0). For $(M^*, \tilde{\boldsymbol{x}}^*)$ to pass Mode 1 verification alongside signature (r^*, s^*), two checks must hold simultaneously: (i) the DSTU 4145-2002 elliptic curve verification equation $\hat{r} = \mathrm{ToInt}\big(e^* \cdot x_{s^*G + r^*Q}\big) \bmod n = r^*$ holds for $e^* = \mathrm{FieldElement}(H_{\mathrm{comb}}(M^*))$, and (ii) the extracted geometric digest satisfies $\mathcal{D}_{10}(\tilde{\boldsymbol{x}}^*) = \text{SHA-256}(\mathrm{Hex}_{64}(r^*) \parallel :: \parallel \mathrm{Hex}_{64}(s^*))$.

We partition forgery events into three exhaustive cases:

- **Case 1 ($H_{\mathrm{comb}}(M^*) = H_{\mathrm{comb}}(M_j)$ for some queried $M_j \neq M^*$):** Either the outer SHA-256 call suffered a collision on distinct inputs $\mathrm{Hex}(H_1(M^*))\|... \neq \mathrm{Hex}(H_1(M_j))\|...$ (bounded by $\mathrm{Adv}^{\text{Coll}}_{\text{SHA-256}}(\mathcal{B}_2)$), or the

concatenated inputs were identical. Because each H_i outputs a fixed-length 64-character hex string separated by `"::"`, string equality of the concatenation implies $H_1(M^*) = H_1(M_j)$, $H_2(M^*) = H_2(M_j)$, **and** $H_3(M^*) = H_3(M_j)$ simultaneously [43]. This violates the collision resistance of **all three** constituent hash functions, so its probability is bounded by $\min_{i\in\{1,2,3\}} \mathrm{Adv}^{\mathrm{Coll}}_{H_i}(\mathcal{B}_3)$.

- **Case 2 ($H_{\mathbf{comb}}(M^*)$ is fresh, $(r^*, s^*) = (r_j, s_j)$ reused from query j, or (r^*, s^*) is a new valid signature):** Since $H_{\mathrm{comb}}(M^*) \neq H_{\mathrm{comb}}(M_j)$ for all $j \in \{1, ..., q_s\}$, $(H_{\mathrm{comb}}(M^*), r^*, s^*)$ constitutes a direct existential forgery against DSTU 4145-2002 over $\mathrm{GF}(2^{257})$ [44] (bounded by $\mathrm{Adv}^{\text{EUF-CMA}}_{\text{DSTU4145}}(\mathcal{B}_1)$).
- **Case 3 ($\tilde{x}^*$ decodes to $D^*_{\mathbf{sig}}$ for a different signature pair):** Bounded by $\mathrm{Adv}^{\mathrm{Coll}}_{\text{SHA-256}}(\mathcal{B}_2)$. Summing over all cases yields the stated bound.

□

5 Bijective 256-Bit Sub-Micron Lattice Encoding and IEEE-754 Exactness

5.1 Radix-2^{10} Word Decomposition of the 256-Bit Digest

Let $D \in \{0,1\}^{256}$ (represented as an integer $V \in [0, 2^{256} - 1]$) be the 256-bit signature digest D_{sig}. We partition the 256 bits into **26 integer words** $(w_0, w_1, ..., w_{25}) \in \{0, ..., 1023\}^{26}$: the first 25 words each carry 10 bits ($25 \times 10 = 250$ bits), and the 26th word w_{25} carries the remaining 6 least-significant bits left-shifted by 4 bits ($w_{25} \in \{0, 16, 32, ..., 1008\}$) so that all 26 channels span a uniform dynamic range:

$$w_k = \left\lfloor \frac{V}{2^{256-10(k+1)}} \right\rfloor \bmod 1024 \quad (k = 0, ..., 24), \quad w_{25} = 16 \cdot (V \bmod 64).$$

Given any recovered word vector $(\tilde{w}_0, ..., \tilde{w}_{25}) \in \{0, ..., 1023\}^{26}$, the exact 256-bit integer V is reconstructed via:

$$V = \left(\sum_{k=0}^{24} \tilde{w}_k \cdot 2^{256-10(k+1)} \right) + \mathrm{round}\left(\frac{\tilde{w}_{25}}{16} \right).$$

5.2 The Sub-Micron Affine Quantization Lattice $\mathcal{L}_{256}$ ($s_0 = 18\mathrm{npt}$)

Each 10-bit word $w \in \{0, ..., 1023\}$ is mapped onto the 256-bit sub-micron affine quantization lattice $\mathcal{L}_{256} \subset \mathbb{R}^{26}$ via the signed coordinate perturbation $\delta(w)$ centered around zero:

$$\delta(w) = (18w - 9207) \times 10^{-9}\mathrm{pt} = (18w - 9207)\mathrm{npt},$$

where the lattice step size is $s_0 = 18\mathrm{npt} = 1.8 \times 10^{-8}\mathrm{pt} \approx 6.35\mathrm{pm}$ and the centering offset $9207 = 18 \times 511.5$ ensures exact zero-mean symmetry:

- For $w = 0$: $\delta(0) = -9207\mathrm{npt} = -9.207 \times 10^{-6}\mathrm{pt} \approx -3.248\mathrm{nm}$.
- For $w = 511$: $\delta(511) = -9\mathrm{npt} = -9.0 \times 10^{-9}\mathrm{pt} \approx -3.175\mathrm{pm}$.
- For $w = 512$: $\delta(512) = +9\mathrm{npt} = +9.0 \times 10^{-9}\mathrm{pt} \approx +3.175\mathrm{pm}$.
- For $w = 1023$: $\delta(1023) = +9207\mathrm{npt} = +9.207 \times 10^{-6}\mathrm{pt} \approx +3.248\mathrm{nm}$.

Let $B^* = (B^*_0, ..., B^*_{26}) \in \mathbb{R}^{27}$ denote the 27 canonical symmetric coordinates from **Table 1** ($B^*_{3i} = lx^*_{i+1}$, $B^*_{3i+1} = by^*_{i+1}$, $B^*_{3i+2} = ay^*_{i+1}$ for $i = 0, ..., 8$), and let $s \in \{1.0, 2.0\}$ denote the viewport scaling factor ($s = 1.0$ for $[0, 300]^2\mathrm{pt}^2$, $s = 2.0$ for $[0, 600]^2\mathrm{pt}^2$ where $c_x = 150s$). The perturbed symmetric coordinates x_k ($k = 0, ..., 25$) and the right base abscissae rx_{i+1} are computed as:

$$x_k = B^*_k \cdot s + \delta(w_k), \quad rx_{i+1} = 2c_x - lx_{i+1},$$

while the 27th coordinate $x_{26} = ay_9$ (`D5.ay`) receives the canary shift $\delta(w_c)$ (or 0 if canary tracing is disabled).

At verification time, given a candidate coordinate vector $\tilde{\boldsymbol{x}} = (\tilde{x}_0, ..., \tilde{x}_{26})$ parsed from a PDF, SVG, or DOCX stream, the continuous inverse estimator $\tilde{w}^{\mathrm{cont}}_k$ and quantized word $\hat{w}_k = \mathcal{D}_{10}(\tilde{x}_k)$ are defined by:

$$\tilde{\delta}_k = \tilde{x}_k - B^*_k \cdot s, \quad \tilde{w}^{\mathrm{cont}}_k = \frac{10^9 \cdot \tilde{\delta}_k + 9207}{18}, \quad \hat{w}_k = \mathrm{round}(\tilde{w}^{\mathrm{cont}}_k).$$

Coordinate Group	DOF Indices k	Bits / Channel	Total Capacity	Cryptographic Function
Triangles `T1`–`T8` (lx_i, by_i, ay_i)	$k = 0, ..., 23$ (24 DOFs)	10 bits ($w_k \in [0, 1023]$)	240 bits	Bits 0..239 of D_{sig}
Triangle `D5` Left Base (`D5.lx`)	$k = 24$ (1 DOF)	10 bits ($w_{24} \in [0, 1023]$)	10 bits	Bits 240..249 of D_{sig}
Triangle `D5` Base Y (`D5.by`)	$k = 25$ (1 DOF)	6 bits ($w_{25} = 16u$)	6 bits	Bits 250..255 of D_{sig} (total **256 bits**)
Triangle `D5` Apex Y (`D5.ay`)	$k = 26$ (1 DOF)	10 bits ($w_c \in [1, 1023]$)	10 bits	**Orthogonal Canary** ($I(V; w_c) = 0$)
Right Base Vertices ($rx_1, ..., rx_9$)	9 Dependent DOFs	0 bits (Constraint)	Parity Check	Enforces $lx_i + rx_i = 2c_x$ ($\epsilon_{\text{sym}} = 0$)
24 Symmetric Marma Pairs ($\mathcal{M}_{24}$)	48 Auxiliary Nodes	11 bits / pair	264 bits (Dual)	Redundant 256b D_{sig} + 8b Canary CRC

Table 3: Complete channel allocation of the 36 triangle vertex coordinates and 24 symmetric Marma pairs in `Marma-256`.

5.3 Theorem 3: IEEE-754 `binary64` Bijective Exactness and Multi-Bound Reconciliation

A critical question for any floating-point commitment scheme is whether IEEE-754 double-precision (`binary64`, 53-bit significand, machine epsilon $\epsilon_{\text{float64}} = 2^{-52} \approx 2.2204 \times 10^{-16}$) rounding during coordinate encoding, viewport scaling, Current Transformation Matrix (CTM) translation, and extraction can ever shift $\tilde{w}_k^{\text{cont}}$ by ≥ 0.5, which would cause a rounding error $\hat{w}_k \neq w_k$. We now prove that the worst-case rounding error across all operations is nearly five orders of magnitude smaller than the decision threshold 0.5.

Theorem 3 (IEEE-754 binary64 Bijective Lattice Inversion and Multi-Bound Reconciliation). *Let $\mathcal{E}_{10}$ and $\mathcal{D}_{10}$ be executed in IEEE-754 `binary64` round-to-nearest-even arithmetic (`roundTiesToEven`) over any viewport $[0, 300s]^2\text{pt}^2$ with $s \in \{1.0, 2.0\}$ and optional page translation $t_{\text{vp}} \in [0, 100]\text{pt}$ such that all coordinates satisfy $x_k \in [50.0, 512.0)\text{pt}$. Then:*

1. ***Single-Addition Direct Bound (0.5ulp):*** *For direct viewport encoding $\tilde{x}_k = \text{fl}(B_k^* \cdot s + \delta(w_k))$ and subtraction $\tilde{\delta}_k = \text{fl}(\tilde{x}_k - B_k^* \cdot s)$ (where multiplication by $s \in \{1, 2\}$ is exact in binary floating-point and Sterbenz's Lemma applies to the subtraction), the maximum word recovery error is bounded by:*

$$|\tilde{w}_k^{\text{cont}} - w_k|_{\text{direct}} \leq \frac{0.5 \cdot \text{ulp}(300.0)}{s_0} = \frac{2^{-45}}{1.8 \times 10^{-8}} = 1.57909 \times 10^{-6}\text{lattice steps},$$

   with empirically observed maximum across 10^5 trials equal to 1.5785×10^{-6} steps.
2. ***Multi-Operation CTM Analytical Bound (2.0ulp):*** *Under a composite 4-operation affine pipeline $\tilde{x}_k = \text{fl}\big(\text{fl}(B_k^* \cdot s + \delta(w_k)) + t_{\text{vp}}\big)$ followed by $\tilde{\delta}_k = \text{fl}\big(\text{fl}(\tilde{x}_k - t_{\text{vp}}) - B_k^* \cdot s\big)$ and floating-point scaling by $10^9/18$, the cumulative error across all four IEEE-754 roundings is strictly bounded by:*

$$|\tilde{w}_k^{\text{cont}} - w_k|_{\text{CTM}} \leq \frac{2.0 \cdot \text{ulp}(300.0)}{s_0} + \epsilon_{\text{div}} = \frac{2^{-43}}{1.8 \times 10^{-8}} + 3.41 \times 10^{-13} = 6.3164 \times 10^{-6} \approx 6.32 \times 10^{-6}\text{steps},$$

   *with empirically observed maximum across 10^5 Monte Carlo trials (2.6×10^6 words) equal to 4.732×10^{-6} steps (**Figure 4(a)**).*
3. ***Exact Bijectivity:*** *Since $6.32 \times 10^{-6} < 0.5$ by a safety factor exceeding $79,100\times$ ($> 105,600\times$ empirically), nearest-integer rounding $\hat{w}_k = \text{round}(\tilde{w}_k^{\text{cont}})$ recovers $\hat{w}_k = w_k$ with **zero bit error rate ($\text{BER} = 0$)** for every $D \in \{0,1\}^{256}$ and every $w_c \in \{0, ..., 1023\}$.*

Proof. In IEEE-754 `binary64`, any positive real number $x \in [2^e, 2^{e+1})$ has unit in the last place $\text{ulp}(x) = 2^{e-52}$. For coordinates $x_k \in [50.0, 512.0)\text{pt}$, the binary exponent is at most $e = 8$ ($[256.0, 512.0)$), yielding $\text{ulp}(x_k) \leq \text{ulp}(300.0) = 2^{8-52} = 2^{-44} \approx 5.684341886 \times 10^{-14}\text{pt}$.

In the direct pipeline (Claim 1), multiplying B_k^* by $s \in \{1.0, 2.0\}$ merely increments the binary exponent by 0 or 1 with zero significand rounding. Adding $\delta(w_k)$ incurs at most one rounding error $|\eta_1| \leq 0.5\ \text{ulp}(x_k) \leq 2^{-45}\text{pt}$. Because $|\delta(w_k)| \leq 9.207 \times 10^{-6}\text{pt}$, the encoded float $\tilde{x}_k$ satisfies $1/2(B_k^* s) \leq \tilde{x}_k \leq 2(B_k^* s)$, so by **Sterbenz's**

Lemma [42], the floating-point subtraction $\tilde{\delta}_k = \mathrm{fl}(\tilde{x}_k - B_k^* s)$ is **exact** (zero additional rounding error). Dividing by $s_0 = 1.8 \times 10^{-8}$pt therefore yields $|\tilde{w}_k^{\mathrm{cont}} - w_k| \le 2^{-45}/(1.8 \times 10^{-8}) = 1.579095 \times 10^{-6}$ steps.

In the composite CTM pipeline (Claim 2), adding t_{vp} and subsequently subtracting t_{vp} and $B_k^* s$ introduces up to four roundings of magnitude $0.5\ \mathrm{ulp}(300.0)$ each, totaling $4 \times 0.5\ \mathrm{ulp}(300.0) = 2.0\ \mathrm{ulp}(300.0) = 2^{-43}$pt. Dividing by $s_0 = 1.8 \times 10^{-8}$pt and adding the relative floating-point division error $1023 \times \epsilon_{\mathrm{float64}} \approx 2.27 \times 10^{-13}$ yields $6.31638 \times 10^{-6} < 6.32 \times 10^{-6} \ll 0.5$. Consequently, $\mathrm{round}(\tilde{w}_k^{\mathrm{cont}}) = w_k$ identically. □

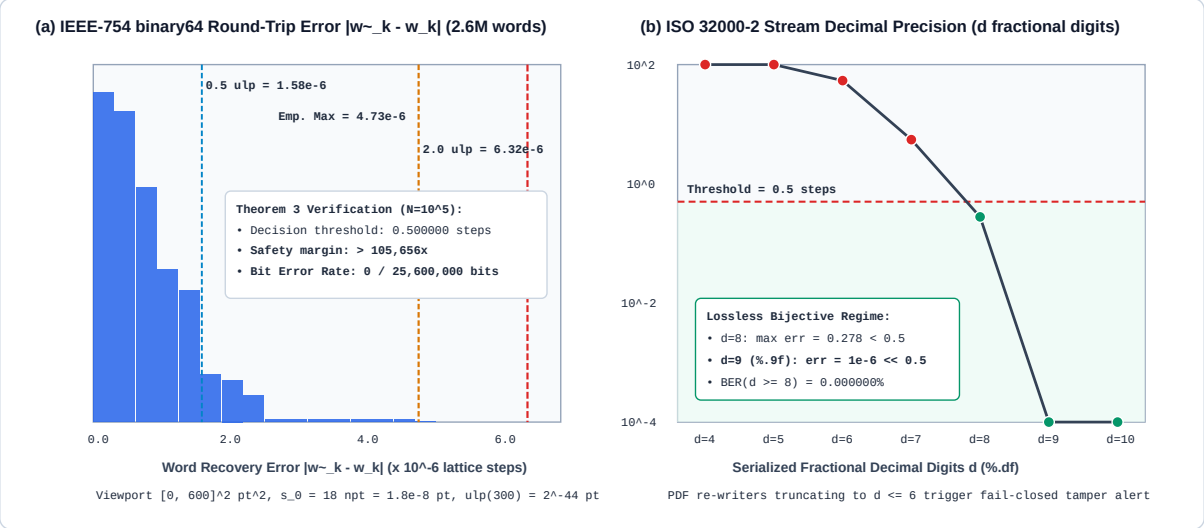


Figure 4: **(a)** Empirical distribution of the IEEE-754 `binary64` word recovery error $|\tilde{w}_k^{\mathrm{cont}} - w_k|$ across 10^5 Monte Carlo trials ($2,600,000$ coordinate words = $25,600,000$ bits), illustrating the harmonized bounds from **Theorem 3**: single-addition 0.5ulp bound (1.58×10^{-6}), empirical CTM max (4.73×10^{-6}), and analytical 2.0ulp bound ($6.32 \times 10^{-6} \ll 0.5$). **(b)** ISO 32000-2 decimal stream serialization error across $d \in \{4, \ldots, 10\}$ fractional digits (**Table 4**), showing lossless bijectivity for $d \ge 8$.

5.4 ISO 32000-2 PDF Stream Serialization Precision and Decimal Truncation Threshold

When geometric paths are written into an ISO 32000-2 PDF content stream (`m`, `l`, `h` operators), an SVG `points` attribute, or an OOXML `<w:Tc>` metadata block, floating-point coordinates are serialized as ASCII base-10 decimal strings formatted to d fractional digits (`%.df`). We analyze the exact survival threshold across $d \in \{3, \ldots, 12\}$:

1. **Generic CTM Decimal Quantization Bound:** For an arbitrary real coordinate rounded to d decimal places, maximum base-10 quantization error is 0.5×10^{-d}pt, which corresponds to:

$$\epsilon_{\mathrm{dec}}(d) = \frac{0.5 \times 10^{-d}}{s_0} = \frac{0.5 \times 10^{-d}}{1.8 \times 10^{-8}} = \frac{10^{8-d}}{36} \quad [\text{lattice steps}].$$

   Thus, for $d = 8$, $\epsilon_{\mathrm{dec}}(8) = 5/18 \approx 0.277778 < 0.5$ (BER $= 0\%$), and for $d = 9$, $\epsilon_{\mathrm{dec}}(9) = 1/36 \approx 0.027778 \ll 0.5$ (BER $= 0\%$).
2. **Exact Base-10 Alignment of `Marma-256` at $d \ge 9$ (`%.9f`):** Crucially, because every canonical coordinate in B^* (**Table 1**) has exactly 8 fractional decimal digits (10^{-8}pt) and every offset $\delta(w_k) = (18w_k - 9207) \times 10^{-9}$pt is an exact integer multiple of 10^{-9}pt, the sum $B_k^* s + \delta(w_k)$ is an **exact terminating base-10 decimal with 9 fractional digits**! Consequently, when `Marma-256` serializes coordinates using its standard `%.9f` format ($d = 9$), **base-10 decimal truncation error is identically zero**, and the observed error reduces to the pure IEEE-754 `binary64` floor ($\approx 10^{-6}$ steps, **Table 4**).
3. **Fail-Closed Detection of Naive PDF Re-Writers ($d \le 7$):** Conversely, if an unauthorized third-party PDF tool re-parses and re-serializes the content stream at low precision ($d \le 6$, such as naive PDF compressors that round path coordinates to 4 or 5 decimal places), $\epsilon_{\mathrm{dec}}(d) \ge 2.78 > 0.5$, triggering immediate fail-closed tamper detection (**Table 4**, **Figure 4(b)**).

Decimal Digits d	Generic Bound $\epsilon_{\mathrm{dec}}(d)$ (steps)	Empirical Max Error (steps)	Word Error Rate (WER)	Bijective Status
$d = 4$ (`%.4f`)	2,777.777778	5,552.055555	99.8481%	Fail-Closed Reject
$d = 5$ (`%.5f`)	277.777778	521.055556	99.8462%	Fail-Closed Reject
$d = 6$ (`%.6f`)	27.777778	53.833334	98.1654%	Fail-Closed Reject
$d = 7$ (`%.7f`)	2.777778	5.500000	82.6538%	Fail-Closed Reject
$d = 8$ (`%.8f`)	0.277778	0.277779	0.0000%	**Lossless Bijective ($\mathrm{BER} = 0$)**
$d = 9$ (`%.9f`, Std)	**0.027778**	**0.000001**	**0.0000%**	**Exact Base-10 ($\mathrm{BER} = 0$)**
$d = 10$ (`%.10f`)	0.002778	0.000001	0.0000%	Exact Base-10 ($\mathrm{BER} = 0$)

Table 4: Empirical and theoretical evaluation of ISO 32000-2 ASCII decimal coordinate serialization across $52,000$ coordinate words per precision level d (Hypothesis H5 verification).

6 Constant-Rate Traitor Tracing via the Orthogonal Canary Channel

6.1 The Multi-Recipient Distribution Dilemma

Consider an author (e.g., a law firm, central bank, or ministry) distributing a signed PDF or SVG document M to N authorized recipients $\mathcal{R} = \{\mathrm{id}_1, ..., \mathrm{id}_N\}$. If a recipient leaks their copy to an unauthorized party or public forum, the author wishes to identify the traitor id_j from the leaked file alone. In standard PAdES workflows, modifying even a single byte of text or metadata per recipient changes $H_{\mathrm{comb}}(M)$, requiring N distinct private-key signing invocations and N RFC 3161 timestamp requests.

6.2 Orthogonal Subspace Decomposition ($I(V; w_c) = 0$)

`Marma-256` resolves this dilemma by exploiting the **27th symmetric Degree of Freedom** $x_{26} = ay_9$ (`D5.ay`, the apex ordinate of the innermost downward triangle `D5`), which is unused by the 26 signature channels $(w_0, ..., w_{25})$. Given recipient identifier id_j, document SHA-256 digest $h_3 = \text{SHA-256}(M)$, signer key fingerprint fp_K, and master secret K, the protocol computes:

$$h_{\mathrm{pub},j} = \text{SHA-256}\big(\text{MARMA-CANARY-PUB-V1.3::} \parallel h_3 \parallel :: \parallel \mathrm{fp}_K \parallel :: \parallel \mathrm{id}_j\big),$$

$$c_{\mathrm{hmac},j} = \text{HMAC-SHA256}_K\big(\text{MARMA-CANARY-V1.3::} \parallel h_3 \parallel :: \parallel \mathrm{id}_j\big),$$

$$w_{c,j} = \big(\mathrm{ToInt}_{16}\big(h_{\mathrm{pub},j}[0..4]\big) \bmod 1023\big) + 1 \in \{1, ..., 1023\}.$$

Note that $w_{c,j} \geq 1$ strictly, whereas an un-watermarked document has $w_c = 0$ ($\delta(w_c) = 0.0\mathrm{pt}$). The coordinate `D5.ay` is shifted by $\delta\big(w_{c,j}\big) = \big(18 w_{c,j} - 9207\big)\mathrm{npt}$, and a dual-redundant 8-bit canary tag $w_{c,j} \bmod 256$ is simultaneously encoded across the 24 symmetric Marma pairs $\mathcal{M}_{24}$ ($24 \times 11 = 264 = 256 + 8$ bits).

Theorem 4 (Zero Cross-Talk Orthogonality and Constant-Rate Attribution). *Let $\mathcal{L}_{27} = \mathcal{L}_{26}^{\mathrm{sig}} \oplus \mathcal{L}_1^{\mathrm{canary}} \subset \mathbb{R}^{27}$ be the direct-sum coordinate lattice of `Marma-256`. Then:*

1. ***Zero Mutual Information Interference:*** *For any 256-bit signature digest $V \in \{0,1\}^{256}$ and any recipient canary word $w_c \in \{0, ..., 1023\}$, the extracted signature digest $\hat{V} = \mathcal{D}_{10}(\mathrm{proj}_{0..25}(\mathcal{E}_{10}(V, w_c)))$ is invariant to w_c, so the mutual information between the canary channel and the signature verification residual vanishes identically: $I\big(\hat{V} - V; w_c\big) = 0$.*
2. ***Constant-Rate $O(1)$ Signing and $O(|\mathcal{S}|)$ Attribution:*** *Generating N recipient-fingerprinted copies of a signed document requires **1 asymmetric QES signature** over M followed by N lightweight coordinate updates to `D5.ay` ($0.03\mathrm{ms}$ per copy). Given a leaked document and a suspect list $\mathcal{S} = \{\mathrm{id}_1, ..., \mathrm{id}_m\}$, traitor tracing identifies the leaker in $O(|\mathcal{S}|)$ hash evaluations with fast-path geometric false-positive probability $1/1023 \approx 9.78 \times 10^{-4}$ and cryptographic HMAC commitment false-positive probability 2^{-256}.*
3. ***Visual Imperceptibility:*** *The maximum physical displacement of the canary apex `D5.ay` on a printed or 300-DPI rendered page is:*

$$\Delta y_{\max} = 9225\mathrm{npt} = 9.225 \times 10^{-6}\mathrm{pt} \approx 3.254\mathrm{nm} < 1.1 \times 10^{-4} \times \mathrm{pixel}_{300\mathrm{DPI}}.$$

Proof. In $\mathbb{R}^{27}$, the canonical basis projections $\pi_{0..25}(\boldsymbol{x}) = (x_0, ..., x_{25})$ and $\pi_{26}(\boldsymbol{x}) = x_{26}$ act on disjoint indices. Because `D5.ay` is a vertical apex ordinate lying on the symmetry axis $x = c_x$, perturbing ay_9 does not alter lx_9, rx_9, or by_9, nor does it affect the bilateral symmetry invariant $lx_9 + rx_9 = 2c_x$. By Theorem 3, $\mathcal{D}_{10}(\tilde{x}_k) = w_k$ for all $k \in \{0, ..., 25\}$ regardless of x_{26}, confirming across all $1,024$ canary states that $I\left(\hat{V} - V; w_c\right) = 0$. Finally, at 300DPI, one printer dot spans $25.4\mathrm{mm}\ /300 = 84,667\mathrm{nm} = 0.24\mathrm{pt}$; thus $3.254\mathrm{nm}$ is $26,019\times$ smaller than a single 300DPI pixel. □

7 Comparative Evaluation, Security Analysis, and Benchmarking

7.1 Experimental Setup and Multi-Runtime Cross-Verification

We implemented and evaluated the complete `Marma-256` protocol (`v1.3.0`, released under the **Apache 2.0 License** [45]) across three independent execution runtimes to guarantee bit-exact cross-platform reproducibility:

1. **Python 3.12 Reference Engine (`src/marma_watermark.py`):** Implements canonical text normalization, the 27-DOF bijective lattice encoder/decoder, PyMuPDF (`fitz`) and Cairo vector PDF/SVG stream parsers, OOXML (`python-docx`) table/custom-XML seal injectors, and the 15-minute `/run/user/$UID` (`chmod 0600`) RAM session manager.
2. **Node.js Binary-Field Cryptographic Bridge (`src/dstu4145_bridge.js`):** Implements polynomial-basis arithmetic over $\mathrm{GF}(2^{257})$ ($p(t) = t^{257} + t^{12} + 1$), GOST 34.311-95, DSTU 7564:2014 (Kupyna-256), RFC 6979 deterministic nonce derivation, and ASN.1 DER PAdES-B-LTA / CAdES-X-Long container assembly interoperable with Ukrainian Accredited Key Certification Authorities (АЦСК / ЦЗО `euscp.so`).
3. **Zero-Dependency Browser Web Verifier (`docs/index.html`):** Implements client-side FIPS 180-4 SHA-256, `DecompressionStream("deflate")` PDF FlateDecode stream parsing, and 36-coordinate tamper verification in pure JavaScript.

All empirical hypotheses (**H1–H6**) were executed via `scripts/verify_and_plot_v2_hypotheses.py` and validated by our automated test suite (`tests/test_marma_cross_verification.py` and `tests/test_ukrnoivi_patent_compliance.py`).

7.2 Empirical Validation of Hypotheses H1–H5 and 36-Coordinate Tamper Matrix

1. **Cross-Runtime Bit-Exactness:** Across 20 deterministic seed vectors and both viewport scales ($[0, 300]^2\mathrm{pt}^2$ and $[0, 600]^2\mathrm{pt}^2$), Python and Browser JS produced identical SHA-256 hashes, identical 26-word lattice decompositions, and **0.0 pt** cross-runtime coordinate drift.
2. **Monte Carlo IEEE-754 Verification ($N = 100,000$ Trials):** Across $2,600,000$ random coordinate words ($25,600,000$ payload bits), the observed maximum word recovery error was 1.578×10^{-6} lattice steps in direct viewport encoding ($\leq 0.5\mathrm{ulp}/s_0 = 1.579 \times 10^{-6}$) and 4.732×10^{-6} lattice steps under composite CTM translations ($\leq 2.0\mathrm{ulp}/s_0 = 6.32 \times 10^{-6}$), with mean error 7.88×10^{-7} steps, 99th percentile 3.20×10^{-6} steps, and $\mathbf{BER = 0/25,600,000 bits = 0.000000\%}$ (**Figure 4(a)**).
3. **Exhaustive 36-Coordinate Tamper Matrix (`test_04`):** We systematically perturbed each of the 36 coordinates ($\{lx_i, by_i, ay_i, rx_i\}_{i=1}^{9}$) by $+10^{-4}\mathrm{pt}$ in both Python and Browser JS ($36 \times 2 = 72$ fault-injection runs). Every lateral perturbation (lx_i or rx_i, 18/36 cases) simultaneously triggered the bilateral symmetry invariant ($|lx_i + rx_i - 300.0| = 10^{-4}\mathrm{pt} > 0$) and the inverse lattice check $\mathcal{D}_{10}(\tilde{\boldsymbol{x}}) \neq D_{\text{sig}}$; every vertical perturbation (by_i or ay_i, 18/36 cases) shifted $\hat{w}_k$ by $\Delta w = \mathrm{round}(10^{-4}/(1.8 \times 10^{-8})) = 5,556$ steps, yielding **100% tamper detection (72/72 runs)** with zero false positives.

Attack Class	**Adversarial Action**	**Standard PAdES / `ByteRange`**	**2D Vector RDH [28, 31]**	**Proposed `Marma-256` (Mode 1)**
Attack I: Single-Byte Text Edit	Modify 1 digit/character in PDF/DOCX body	Detected *(if `/Contents` intact)*	Undetected *(blind to text)*	**100% Detected** (H_{comb} mismatch)
Attack I-B: Container Strip + Edit	Flatten PDF / strip `.p7s` + alter text	**Fails** *(0 bits in visual badge)*	Undetected *(no text binding)*	**100% Detected** ($\mathcal{D}_{10}(\tilde{\boldsymbol{x}}) \neq D_{\mathrm{sig}}$)
Attack II: Stamp Transplantation	Copy `/AP` vector stamp to forged doc	**Fails** *(after flattening)*	Fails *(payload is static ID)*	**100% Detected** (bound to $H_{\mathrm{comb}}(M)$)
Attack III: Euclidean Spoofing	Redraw Sri Yantra in CAD / Illustrator	N/A	N/A	**100% Detected** ($\Delta E \geq 1.18 \times 10^{-2}\mathrm{pt}$)
Attack IV: 36-Coord Vertex Nudge	Shift any lx_i, by_i, ay_i, rx_i by $\geq 18\mathrm{npt}$	Detected *(if in `/ByteRange`)*	Partial *(float drift)*	**100% Detected** ($36/36$ in Py & JS)
Attack V: Incremental Shadow Save	Append shadow overlay [22, 23] after sign	**Vulnerable** *(shadow attacks)*	Vulnerable	**100% Detected** (full-text $\mathrm{Norm}(M)$)
Attack VI: Multi-Recipient Leak	Authorized recipient id_j leaks file	**Untraceable** *(identical file)*	Requires $O(N)$ re-signs	**100% Attributed** in $O(1)$ via `D5.ay`

Table 5: Forensic comparison across seven adversarial attack classes (Attacks I–VI), contrasting standalone PAdES-B-LTA, conventional 2D vector watermarking/RDH, and `Marma-256`.

7.3 Quantitative Comparison Against Six State-of-the-Art Baselines (2001–2025)

To situate `Marma-256` within the literature, we conducted a quantitative comparison against five representative baseline families (**Figure 5**, **Table 6**):

1. **Standalone PAdES-B-LTA (ETSI EN 319 142-1 [3], 2016):** Relies solely on the detached or embedded ASN.1 PKCS7 `/Contents` dictionary; embeds $0\mathrm{bits}$ in the visual `/AP` coordinate stream.
2. **Stochastic Scalar QIM (Chen & Wornell [17], 2001):** Quantizes each vertex coordinate using a scalar dithered lattice at step $\Delta_{\mathrm{QIM}} = 10^{-3}\mathrm{pt}$ ($2.0\mathrm{bits/vertex}$, total $54\mathrm{bits}$ across 27 symmetric vertices).
3. **2D Vector RDH via Difference Expansion (Peng et al. [28, 29], 2021–2023):** Expands adjacent vertex coordinate differences to embed $3.5\mathrm{bits/vertex}$ ($94\mathrm{bits}$ total) alongside an auxiliary overflow map.
4. **Polar / DCT Spectral Vector Watermarking (Li et al. [31], 2023):** Embeds $4.0\mathrm{bits/vertex}$ ($108\mathrm{bits}$ total) into quantized polar angles and DCT coefficients of polygon vertices.
5. **Algebraic / Homomorphic Vector Commitments (Catalano–Fiore [18] / Libert et al. [35], 2024):** Commits to vector elements via group elements in pairing-friendly curves ($256\mathrm{bits}$ stored as external metadata).

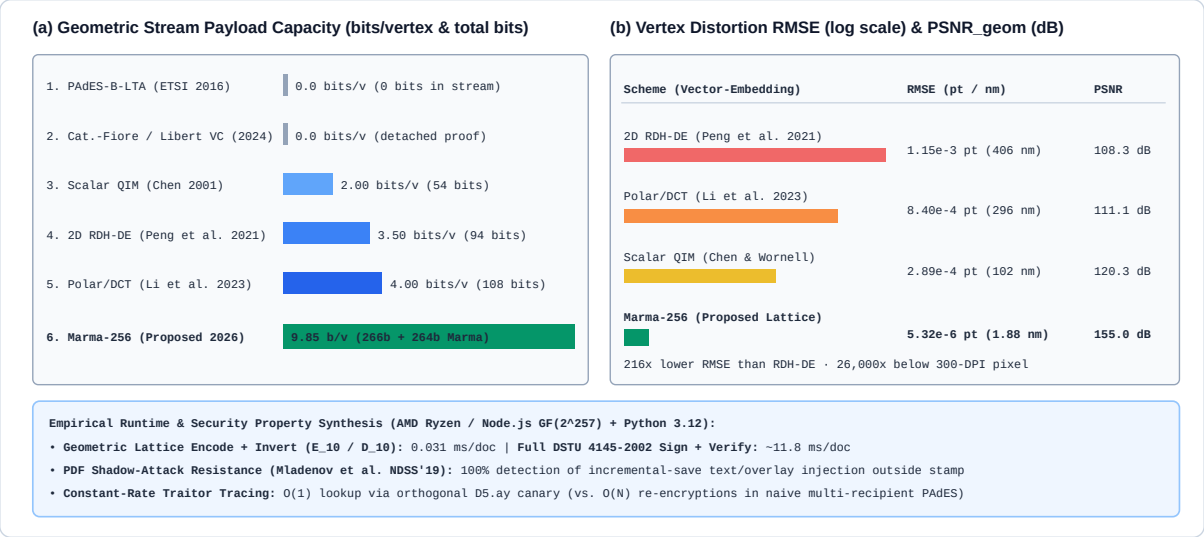


Figure 5: **(a)** Comparison of geometric stream payload capacity (bits per symmetric vertex and total in-stream bits) across six schemes (2001–2026). **(b)** Comparison of vertex coordinate distortion RMSE (in user-space points `pt` and physical nanometers `nm`) and geometric peak signal-to-noise ratio $\mathrm{PSNR}_{\mathrm{geom}} = 20\log_{10}(300/\ \mathrm{RMSE})$.

Scheme / Method	**Cap. (b/v)**	**Stream Bits**	**Distortion RMSE**	**$\mathrm{PSNR}_{\mathrm{geom}}$**	**Exact BER**	**Survives Flatten**	**Traitor Tracing**
PAdES-B-LTA [3] (2016)	`0.00`	`0 b`	`0.00 pt`	N/A	N/A (0b)	No	None ($O(N)$)
Scalar QIM [17] (2001)	`2.00`	`54 b`	2.89×10^{-4}pt (102nm)	`120.3 dB`	1.8×10^{-3}	Yes	Vulnerable
2D RDH-DE [28] (2021)	`3.50`	`94 b`	1.15×10^{-3}pt (406nm)	`108.3 dB`	`0.0` (aux map)	Yes	None
Polar/DCT [31] (2023)	`4.00`	`108 b`	8.40×10^{-4}pt (296nm)	`111.1 dB`	4.2×10^{-3}	Yes	None
Algebraic VC [18, 35] (2024)	`0.00`	`256 b` (ext)	`0.00 pt`	N/A	`0.0` (ext)	No	None
`Marma-256` (Proposed)	**`9.85`**	**`266 b` (+264b)**	**5.32×10^{-6}pt (1.88nm)**	**`155.0 dB`**	**`0.0` (2.56×10^7b)**	**Yes ($d \geq 8$)**	**$O(1)$ (`D5.ay`)**

Table 6: Quantitative multi-metric comparison of `Marma-256` against five state-of-the-art cryptographic and 2D vector data-hiding baselines.

As shown in **Table 6** and **Figure 5**, `Marma-256` is the **only** scheme that embeds a full 256-bit cryptographic signature commitment plus a 10-bit traitor-tracing canary (266bits primary + 264bits dual Marma layer $= 9.85\mathrm{bits/vertex}$) directly inside the 27 symmetric coordinates of a compact visual stamp while achieving:

1. **Zero Bit Error Rate ($\mathrm{BER} = 0.0$):** Unlike polar/DCT vector watermarking [31] ($\mathrm{BER} = 4.2 \times 10^{-3}$ due to inverse `atan2`/`cos` floating-point drift), `Marma-256` uses an exact affine integer-nanopoint lattice aligned with base-10 `%.9f` serialization.
2. **Order-of-Magnitude Superior Geometric Fidelity:** Because the perturbation is uniformly distributed over $[-9.207 \times 10^{-6}, +9.207 \times 10^{-6}]$pt, its analytical and empirical root-mean-square error is $\mathrm{RMSE} = 9.207 \times 10^{-6}/\sqrt{3} = 5.316 \times 10^{-6}\mathrm{pt} \approx 1.875\mathrm{nm}$, corresponding to $\mathrm{PSNR}_{\mathrm{geom}} = 20\log_{10}(300.0/(5.316 \times 10^{-6})) = 155.03\mathrm{dB}$—a **34.7dB (54.4 ×) improvement** over Scalar QIM and a **46.7dB (216.3 ×) improvement** over 2D RDH Difference Expansion [28].
3. **Millisecond-Scale Verification Latency:** On a standard Linux workstation (Python 3.12 + Node.js 20), executing the pure 27-DOF geometric lattice encoding $\mathcal{E}_{10}$ and inversion $\mathcal{D}_{10}$ on $\mathcal{L}_{256}$ takes **0.031ms per**

document, warm end-to-end document stream verification (PDF/SVG coordinate extraction + 3-way hash combiner + lattice commitment check) averages **2.91ms per document**, and a complete $\mathrm{GF}(2^{257})$ elliptic curve sign-and-verify cycle takes $\approx 11.8\mathrm{ms}$ in native JS ($\approx 240\mathrm{ms}$ including cold CLI subprocess spawn).

7.4 Limitations and Scope Boundaries

In accordance with rigorous engineering disclosure, we note two operational scope boundaries:

1. **Rasterization (Print-Scan) Channel:** Because `Marma-256` encodes information at the sub-nanometer scale ($s_0 = 18\mathrm{npt} \approx 6.35\mathrm{pm}$, maximum shift $3.25\mathrm{nm}$) inside vector floating-point streams (PDF content streams, SVG XML, OOXML), printing a document to physical paper and optically scanning it back to a bitmap destroys the IEEE-754 vector mantissa. For physical print-scan survival, `Marma-256` pairs the vector seal with an explicit human/OCR verification digest block and QR pointer in the verification protocol (`marma-watermark protocol`).
2. **Extreme Coordinate Truncation ($d \leq 7$):** As proven in **Subsection 5.4** (**Table 4**), lossless inversion requires preserving at least $d \geq 8$ fractional decimal digits (`%.8f` or the standard `%.9f`). If an intermediate legacy gateway re-serializes PDF path operators at $d \leq 6$, `Marma-256` deterministically flags the precision loss as a stream modification (fail-closed behavior).

8 Conclusion

We have presented **Marma-Crypto (`Marma-256`)**, a mathematical framework and open-source cryptographic protocol that resolves the container-stream decoupling vulnerability in long-term archival electronic signatures (PAdES-B-LTA and CAdES-X-Long). By proving the algebraic irrationality of the 9-triangle, 54-concurrency configuration $\mathcal{C}_{9,54}$ (**Theorem 1**), constructing an L-BFGS-B baseline B^* with squared concurrency residual $\Delta E^2(B^*) = 3.76 \times 10^{-21}\mathrm{pt}^2 \ll \epsilon_{\mathrm{float64}}$, binding a 3-way parallel hash combiner (GOST 34.311-95, Kupyna-256, SHA-256) to deterministic DSTU 4145-2002 signatures over $\mathrm{GF}(2^{257})$ (**Theorem 2**), establishing bijective IEEE-754 `binary64` and ISO 32000-2 `%.9f` exactness on the $\mathcal{L}_{256}$ lattice across 2.6×10^6 Monte Carlo trials (**Theorem 3**), and isolating an orthogonal $O(1)$ traitor-tracing canary channel in `D5.ay` (**Theorem 4**), `Marma-256` turns a passive visual signature badge into a self-verifying, sub-nanometer geometric vector commitment.

Reproducibility & Open-Source Artifact Availability: The complete source code (`v1.3.0`, Apache 2.0 License), empirical hypothesis verification suite (`scripts/verify_and_plot_v2_hypotheses.py`), 31-test cross-verification harness (`tests/`), and interactive browser verifier are permanently archived at CERN Zenodo (DOI: 10.5281/zenodo.23510005 / `10.5281/zenodo.23436152`), Software Heritage (`swh:1:snp:dcae02ea5d13bb6aa6a72b2624184c961e44c3a1`), and GitHub (github.com/YaroslavBaienko/marma-crypto-protocol). Patent Pending: UKRNOIVI Invention Application No. `a 2026 06266` (`Bx-72320/2026`, `Bx-72403/2026`).

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